

Cryptography - 2_12

Reminder of the theorem we've seen last time:

SS + HVZK $\implies$ Passive security

We further need one assumption:

- $|\mathcal{B}_{\lambda, pk}|$ is large enough (e.g. $w(\log \lambda)$)

Proof

Let $G_0(\lambda) \equiv G_{\Pi, \mathcal{A}}^{id}(\lambda)$ the game for passive security of Π . Consider $G_1(\lambda)$ identical to $G_0(\lambda)$, but where we use $Sim(pk)$ (where the use of a simulator is granted by HVZK property) to answer the transcript queries in the game $G_0(\lambda)$.

Lemma

$$G_0(\lambda) \approx_c G_1(\lambda)$$

Proof

We will use the HVZK property. It requires a hybrid argument, along these lines:
Given $q = \# \text{queries}$

$$G_0(\lambda) : \mathcal{P}(pk, sk) \rightleftharpoons \mathcal{V}(pk), \dots, \mathcal{P}(pk, sk) \rightleftharpoons \mathcal{V}(pk)$$

$$H^1(\lambda) : Sim(pk), \mathcal{P} \rightleftharpoons \mathcal{V}, \dots, \mathcal{P} \rightleftharpoons \mathcal{V}$$

$\vdots$

$$H^i(\lambda) : \underbrace{Sim(pk), \dots, Sim(pk)}_{\#i}, \underbrace{\mathcal{P} \rightleftharpoons \mathcal{V}, \dots, \mathcal{P} \rightleftharpoons \mathcal{V}}_{q-i}$$

$\vdots$

$$G_1(\lambda) : Sim(pk), \dots, Sim(pk)$$

$$\forall i : H^i(\lambda) \approx_c H^{i+1}(\lambda)$$

Diagram illustrating the state transitions and data flow for the $\mathcal{A}$ and $\mathcal{A}_{HVAK}$ components.

Component $\mathcal{A}$:

- Input: $\#9$
- Internal State: z_i
- Output: pk
- Label: "EMPTY"

Component $\mathcal{A}_{HVAK}$:

- Input: pk, sk, τ
- Output: pk, sk, τ

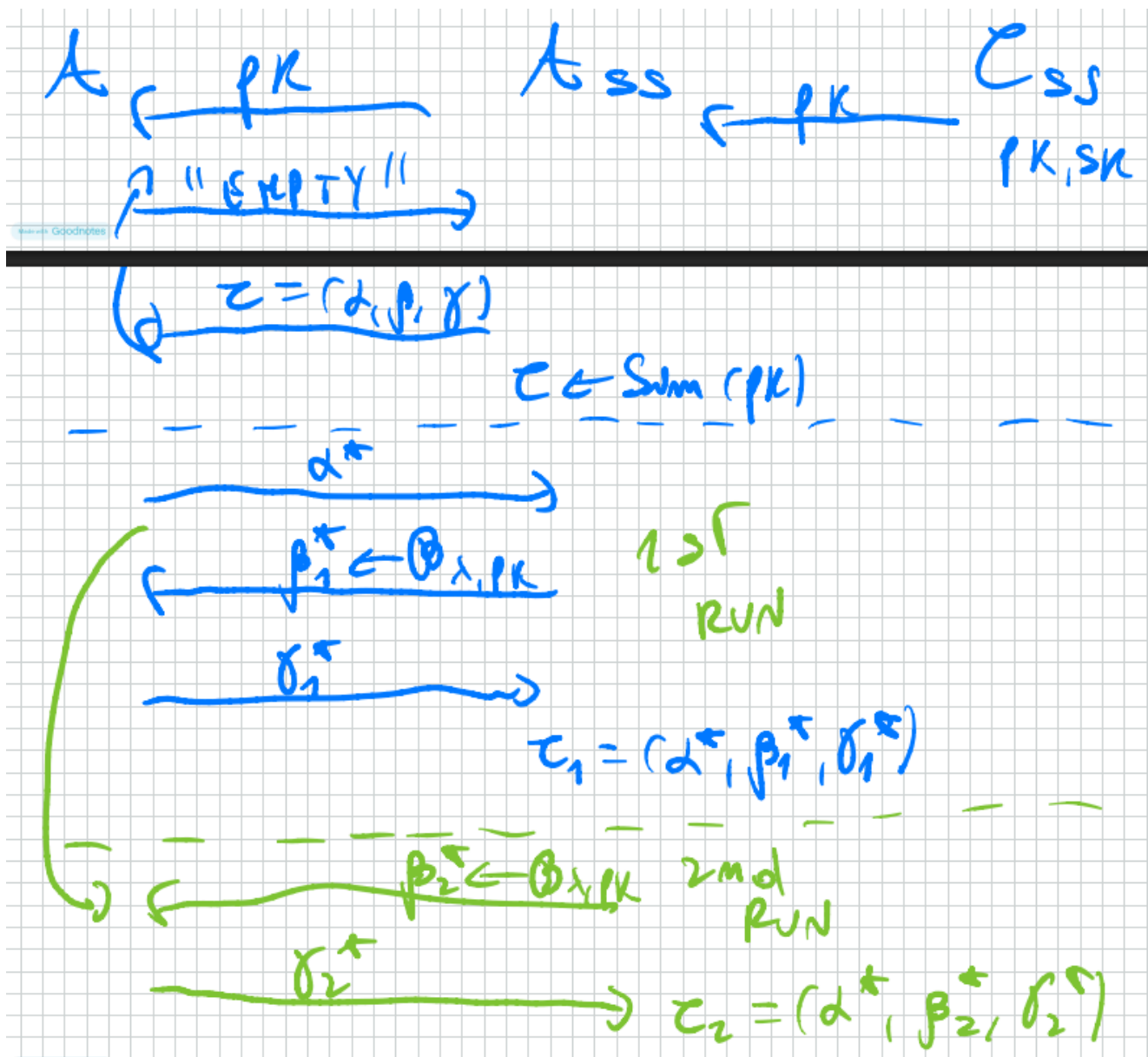
Component $\mathcal{C}_{HVAK}$:

- Input: pk, sk, τ
- Output: $\tau \rightarrow \mathcal{P}_{EV}$
- Output: $\tau \rightarrow \text{Sum}$

Definition of z_i :

$$z_i = \begin{cases} \text{Sum}(pk) & , i \leq n \\ \tau & , i = n+1 \\ \mathcal{P}_{EV} & , i \geq n+2 \end{cases}$$
$$Pr[G_1(\lambda) = 1] \leq \text{negl}(\lambda)$$

$\exists$ PPT $\mathcal{A}$ such that the above is $\geq \frac{1}{poly(\lambda)}$. We build $\mathcal{A}_{SS}$ against SS .



In this reduction, we have A_{SS} trying to win the SS game, and to do this he has to give to the challenger two values of τ such that the verifier approves the exchange. We build A to simulate an adversary able to break passive security, thus able to obtain the secret key.

We have a first phase of A obtaining τ 's generated by A_{SS} , masking as its challenger.

Then, we have a first run in which A starts a "transaction", so he sends α , and in response A_{SS} outputs β from the beta space $(\mathcal{B}_{\lambda, pk})$, and in response we receive γ_1 , computing the first transcript $\tau_1 = (\alpha^*, \beta_1^*, \gamma_1^*)$. Then, because we are in special soundness, we use the same α , getting a new β_2 in the beta space, and we receive γ_2 , thus building a new transcript $\tau_2 = (\alpha^*, \beta_2^*, \gamma_2^*)$. Then A_{SS} sends these two to the challenger.

To analyze this reduction, let $z \in \{0, 1\}^*$ be the RV representing the state of $\mathcal{A}$ after it sent α^* . Then we define:

$$\delta_z = \Pr[G_1(\lambda) = 1 | Z = z]$$

This means: $\Pr[G_1(\lambda) = 1] = \sum_z p_z \cdot \delta_z = E[\delta_z] = \mathcal{E}(\lambda)$ where $p_z = \Pr[Z = z]$.

Finally, let Good be the event that $\beta_1^* \neq \beta_2^*$. Now:

$$\begin{aligned} \Pr[G_{\Pi, A_{ss}}^{SS}(\lambda) = 1] &\geq \Pr[G_{\Pi, A_{ss}}^{SS}(\lambda) = 1 \wedge \text{Good}] \\ &= \Pr[\text{Good}] \cdot \Pr[A_{ss} \text{ wins} | \text{Good}] = (1 - \Pr[\bar{\text{Good}}]) \cdot \Pr[A_{ss} \text{ wins} | \text{Good}] \\ &= \left(1 - \frac{1}{|\mathcal{B}_{\lambda, pk}|}\right) \cdot \Pr[A_{ss} \text{ wins} | \text{Good}] \geq \Pr[A_{ss} \text{ wins} | \text{Good}] - \frac{1}{|\mathcal{B}_{\lambda, pk}|} \\ &\geq \Pr[A_{ss} \text{ wins} | \text{Good}] - \text{negl}(\lambda) = \\ &= \sum_z p_z \delta_z^2 - \text{negl}(\lambda) = E[\delta_z^2] - \text{negl}(\lambda) \\ &\geq (E[\delta_z])^2 - \text{negl}(\lambda) \geq \mathcal{E}^2(\lambda) - \text{negl}(\lambda) \geq \frac{1}{\text{poly}} \end{aligned}$$

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Let's go back to Fiat-Shamir. Given $\Pi = (KGen, \mathcal{P}, \mathcal{V})$ we construct $\Sigma = (Kgen, Sign, Vrfy)$:

- $KGen(1^\lambda) : pk, sk$
- $Sign(sk, m) : \sigma = (\alpha, \gamma)$ such that $\alpha \leftarrow \mathcal{P}_1(pk, sk); \beta = H(\alpha, m); \gamma \leftarrow \mathcal{P}_2(pk, sk, \alpha, \beta)$
- $Vrfy(pk, \sigma, m)$: Output 1 $\iff \mathcal{V}(pk, \alpha, \beta, \gamma) = 1$ with $\beta = H(\alpha, m)$

✓ Theorem

Assuming Π is non-trivial and passively secure, then Σ is UFCMA in the ROM.

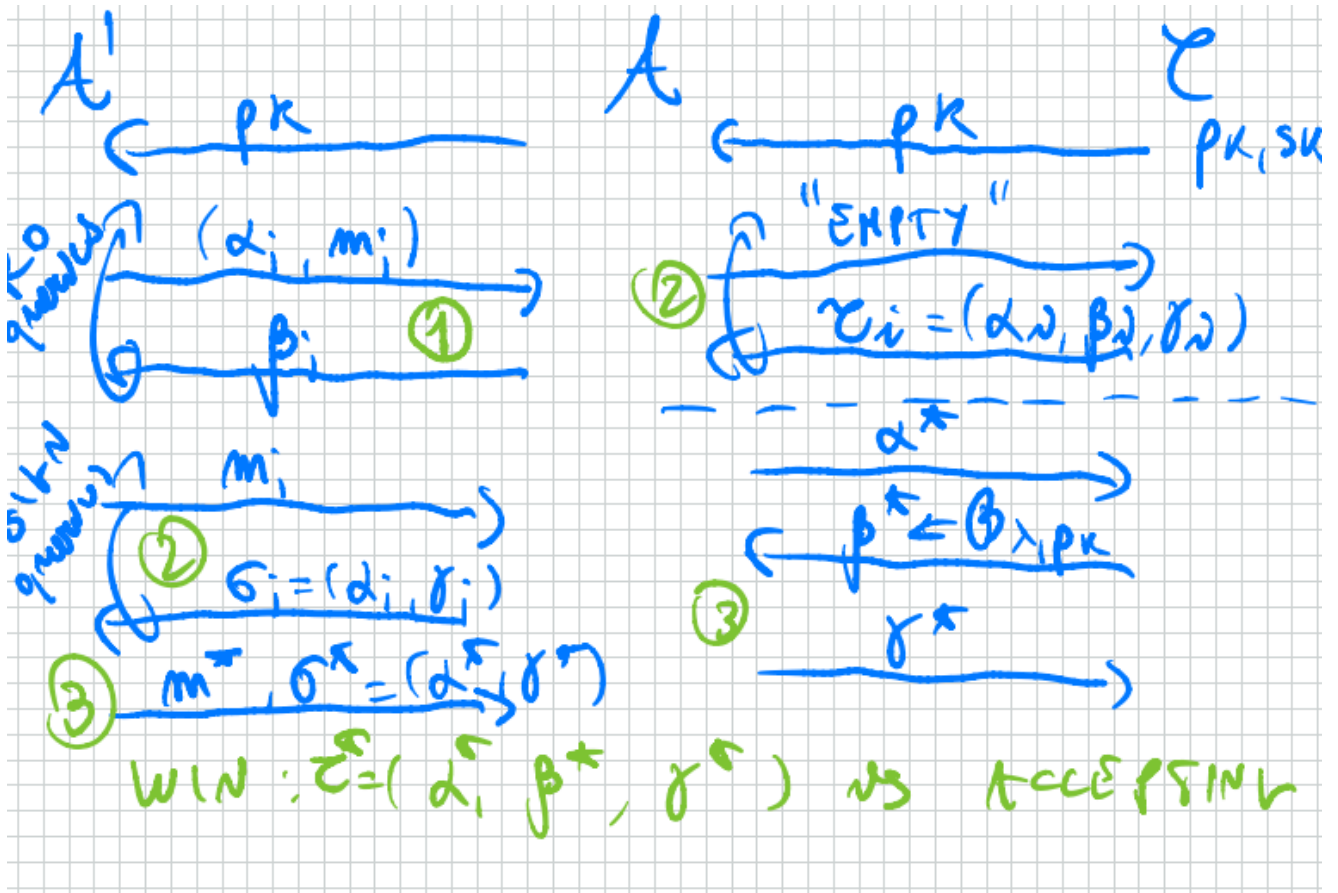
📖 Proof

Let $\mathcal{A}'$ be a PPT adversary breaking UFCMA of Σ w.p. $\frac{1}{\text{poly}(\lambda)}$. Without loss of generality we make a few assumptions on $\mathcal{A}'$:

- It never repeats queries.
- Because each signature query m defines $\sigma = (\alpha, \gamma)$ and $\beta = H(\alpha, m)$, we assume $\mathcal{A}$ never queries (α, m) to the RO after receiving σ .
- Before outputting $\sigma^* = (\alpha^*, \gamma^*)$ it queried (α^*, m^*) to the RO.

Let $q_s = \text{poly}(\lambda) = \# \text{signature queries}$ and $q_h = \text{poly}(\lambda) = \# \text{RO queries}$.

We build $\mathcal{A}$ breaking passive security.



We have $\mathcal{A}'$ sending to $\mathcal{A}$ some RO queries, and $\mathcal{A}'$ sends an empty request to $\mathcal{C}$ receiving valid tags $\tau_i = (\alpha_i, \beta_i, \gamma_i)$, from where $\mathcal{A}$ gets the value of β he needs to send to $\mathcal{A}'$.

Then we have signature queries, where $\mathcal{A}'$ sends m_i , and receives $\sigma_i = (\alpha_i, \gamma_i)$. After all of this, $\mathcal{A}'$ will send $m^*, \sigma^* = (\alpha^*, \gamma^*)$, that will be used by $\mathcal{A}$ for the Σ -protocol with $\mathcal{C}$.

$\mathcal{A}$ also samples $i^* \leftarrow [q_h]$ as the guess for the RO query (α^*, m^*) .

Next:

- Upon input a RO query (m_i, α_i) :
 - If $j \neq i^*$, let $\beta_i \leftarrow \mathcal{B}_{\lambda, pk}$
 - If $j = i^*$, then the reduction hopes $(m_i, \alpha_i) = (m^*, \alpha^*)$. Then initiate the impersonation (3) by sending α^* to $\mathcal{C}$. Let β^* be the challenge from $\mathcal{C}$. Reply to the RO query with β^* . (Then, pause the impersonation).
- Upon input a sign query m_i , output $\sigma_i = (\alpha_i, \gamma_i)$ where γ_i, α_i are from $\tau_i = (\alpha_i, \gamma_i)$ where γ_i, α_i are from $\tau_i = (\alpha_i, \beta_i, \gamma_i)$. Also, program the RO such that $H(\alpha_i, m_i) = \beta_i$.
If (α_i, m_i) was queried before to the RO, then ABORT.
- When $\mathcal{A}'$ outputs $m^*, \sigma^* = (\alpha^*, \gamma^*)$ check that the guess on i^* was correct. If not, abort.
If yes, resume (3) and send γ^* to $\mathcal{C}$.

Analysis:

- If it doesn't abort, the reduction is perfect.
- Probability of not aborting in step (3) is $\frac{1}{\text{poly}(\lambda)} = \frac{1}{q_h}$.
- Probability of aborting in step (2) is negligible. This is because for each signature query $\alpha_i = \alpha_j$ from a previous RO query with $\text{negl}(\lambda)$ probability by non-triviality of Π .
- Probability of not aborting in step (2) is $(1 - q_s \cdot \text{negl}(\lambda)) \geq \frac{1}{\text{poly}(\lambda)}$

$$\implies \Pr[\mathcal{A} \text{ wins}] \geq \frac{1}{\text{poly}(\lambda)} \cdot \frac{1}{?}$$

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